

TECHNICAL DOCUMENT

Lead Conversion

Lightning Web Component and Apex controller that convert a custom Lead__c into an Account and a Contact

Component name	leadConversion (LWC)
Apex class	LeadConvertCls
Test class	LeadConvertClsTest
LWC API version	61.0
Targets	App Page, Home Page, Record Page, Record Action, Flow Screen
Static resource	img (image shown after conversion)
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1. Purpose

The component converts a record of the custom object **Lead__c** into an **Account** and a **Contact**. It replaces the standard Salesforce Lead Convert for this org, because leads are stored in a custom object. It is opened from a Lead record (it listens for the `closequickaction` event, so it is used as a quick action) and walks the user through one screen: confirm the names, click Convert, and see the created records.

Sample Request and Quote creation, pricebook assignment and the "Assigned to Dealer" logic were part of earlier versions. They are now **commented out** in the Apex, LWC, HTML and test code. Only Account and Contact are created today.

1.1 What the user sees

- **Lead has a real "new" flow:** an Account section (Account Name, prefilled from `Company__c`) and a Contact section (Contact Name, prefilled from First and Last name).
- **Lead linked to an existing Account:** an information box explains that a new Contact will be created against that Account and no new Account will be created.
- **Load error box** ("Unable to Load Lead") when the initial load fails, for example when the lead is already converted. The Convert button is disabled.
- **Result screen** after success: "Your Lead Has Been Converted", an image, an Account card (Name, Phone, Email) and a Contact card (Name, Mobile, Email), with **Close** and **Go To Record** buttons. Go To Record opens the Account.

2. Components

Component	Type	Role
leadConversion	LWC (html, js, css, js-meta.xml)	Conversion screen, validation, result display, navigation
LeadConvertCls	Apex class (no sharing keyword)	Loads the lead, creates the Account and Contact, marks the lead as converted, returns the result
LeadConvertClsTest	Apex test	Sixteen tests covering lead load, both conversion paths, field and address mapping, errors and a repeated-conversion loop
img	Static resource	Image shown on the success screen

3. How It Works

1. **Load.** `connectedCallback` calls `LeadConvertCls.getLeadData(recordId)`. Apex queries the lead and throws an error if `Is_Converted__c` is already true. On failure, the LWC shows the Load Error box and a toast.
2. **Decide the path.** The lead formula field `IsNew_Account__c` controls the screen. If true, the Account and Contact name inputs are shown. If false, the existing-Account message is shown. (Per the test class, the formula is true when `Existing_Account__c` points to a placeholder Account named **NEW**; see section 8.)
3. **Validate.** On Convert, the LWC checks that the Account Name and Contact Name are not blank (new path), or that `Existing_Account__c` is set (existing path).
4. **Convert.** The LWC calls `LeadConvertCls.convertLead(accName, conName, leadId)`.

5. **Apex, new Account path.** Creates an Account from the lead (section 5.3) and a Contact from the lead (section 5.4), re-queries both, and sets `Is_Converted__c = true` on the lead.
6. **Apex, existing Account path.** Requires `Existing_Account__c`, loads that Account without changing it, creates a Contact against it, and sets `Is_Converted__c = true`.
7. **Safety.** The method sets a savepoint. Any exception rolls back everything and returns a result with `errorMessage` instead of throwing. DML runs with duplicate rules allowed to save (section 5.2).
8. **Result.** The LWC shows a toast, the Account and Contact cards, and enables Go To Record.

4. LWC Reference

4.1 Files

File	Description
leadConversion.html	Header, load-error box, Account and Contact input sections, existing-account message, result cards, and a fixed footer with Close, Convert and Go To Record. Large commented-out blocks for the old Sample/Quote section.
leadConversion.js	State, Apex calls, validation, navigation. Imports <code>getAccount</code> and <code>getContact</code> but does not use them (their fallback code is commented out).
leadConversion.css	Header, footer, buttons (<code>button-9</code> , <code>button-21</code>) and card styles. Several classes are no longer used.
leadConversion.js-meta.xml	API 61.0; exposed on App Page, Home Page, Record Page, Record Action and Flow Screen; Large and Small form factors for Record Page.

4.2 Main members

Member	Description
@api recordId	The <code>Lead__c</code> Id
connectedCallback	Loads the lead; sets <code>accName</code> (from <code>Company__c</code>), <code>conName</code> (First + Last name), <code>existingAccountId</code> , <code>isNewAccountCheckbox</code> , <code>loadError</code>
handleAcc / handleCon	Store the typed Account and Contact names
handleClick	Validation, then the <code>convertLead</code> call; handles <code>errorMessage</code> in the result and Apex errors
shouldShowAccountSection	True when <code>IsNew_Account__c</code> is true
isConvertDisabled	True while converting or after a load error
onclickGO	Navigates to the created or used Account (<code>standard__recordPage</code>)
handleCloseClick	Dispatches <code>closequickaction</code>
handleCloseOppClick	Dispatches <code>closequickaction</code> , then reloads the page
openAcc / openCon	Expand and collapse the Account and Contact sections

5. Apex Reference: LeadConvertCls

5.1 Methods

Method	Used by LWC	Description
getLeadData(String recId)	Yes	Returns the Lead__c with name, owner, company, status, contact and address fields, Existing_Account__c, IsNew_Account__c and Is_Converted__c. Throws AuraHandledException if already converted.
convertLead(String accName, String conName, Id leadId)	Yes	Main conversion. Returns ConversionResult .
getAccount(String accId, String leadId)	No	Returns an Account with a wide field list (uses the standard Billing/Shipping fields). Only referenced in commented-out LWC code.
getContact(String contactId)	No	Returns a Contact. Only referenced in commented-out LWC code.
getDuplicateBypassDmlOptions() (private)	n/a	See 5.2

ConversionResult (inner class, all **@AuraEnabled**): **accountId**, **contactId**, **errorMessage**, **isNewAccount**, **accountData**, **contactData**, **ownerName**. The old **sampleRequestId**, **quoteId**, **requirementType** and related members are commented out.

5.2 Duplicate rule handling

getDuplicateBypassDmlOptions sets **OptAllOrNone = true**, **DuplicateRuleHeader.AllowSave = true** and **RunAsCurrentUser = true**. Accounts and Contacts are inserted with these options, so duplicate rules (for example fuzzy Account Name matching) do **not** block the conversion. All other DML errors, such as required fields or validation rules, still fail and roll back.

5.3 Lead__c to Account mapping (new Account path)

Lead__c field	Account field
accName (typed by user)	Name
Phone__c, Email__c, Website__c	Phone, Email__c, Website
Customer_Classification__c, Dealer_Assigned__c, Division__c, OEM__c, New_OEM__c	Same names
Type__c	Type
GST_Number__c	GST_Number__c and Shipping_GST_Number__c
PAN_Number__c	PAN_Number__c and Shipping_PAN_Number__c
CIN_Number__c	CIN_Number__c
Id	Lead_Name__c and LeadName__c
LeadSource__c, Subject__c	Lead_Source__c, Lead_Subject__c
Latest_Note__c, Note_Date__c	Latest_Note__c, Note_Date__c
CreatedById, CreatedDate (date part)	Lead_Created_By__c, Lead_Creation_Date__c

Lead__c field	Account field
Industry_Type__c, IndustryMulti__c, Industry_SubType__c	Same names (Industry__c and Industry_Sub_Type__c mappings are commented out)
No_of_Employees__c	NumberOfEmployees (converted to Integer)
Address1__c, City__c, State__c, Country__c, Zip_Postal_Code__c, Other_City__c	Billing_Address__c, Billing_City__c, Billing_State__c, Billing_Country__c, Billing_Zip_Postal_Code__c, Other_Billing_City__c, and the same set with the Shipping_ prefix

5.4 Lead__c to Contact mapping (both paths)

Lead__c field	Contact field
Title__c	Salutation
First_Name__c, Last_Name__c	FirstName, LastName
Designation__c	Title
Email__c, Phone__c, Mobile__c	Email, Phone, MobilePhone
Department__c	Department__c
(new or existing Account)	AccountId

Important: the `conName` value typed by the user is passed to `convertLead` but is **not used**. The Contact name always comes from `First_Name__c` and `Last_Name__c` on the lead.

5.5 Error handling

- A savepoint is set at the start. `DmlException` and any other `Exception` roll back all changes and return a `ConversionResult` with `errorMessage` set (the method does not throw).
- Errors raised inside the method, such as "already converted" or "Existing Account must be selected", are also caught by the general handler, so they reach the LWC as `errorMessage`.
- The LWC shows `errorMessage` in a "Conversion Error" toast.

6. Data Dependencies

Object	Fields and notes
Lead__c	Name, Name__c, OwnerId, Company__c, Lead_Status__c, Lead_Name__c, Phone__c, Mobile__c, Email__c, Website__c, First_Name__c, Last_Name__c, Title__c, Designation__c, Department__c, Existing_Account__c, IsNew_Account__c (formula), Is_Converted__c, Customer_Classification__c, Dealer_Assigned__c, Division__c, GST_Number__c, PAN_Number__c, CIN_Number__c, OEM__c, New_OEM__c, Type__c, Project__c, LeadSource__c, Subject__c, Latest_Note__c, Note_Date__c, Industry__c, Industry_Type__c, Industry_Sub_Type__c, IndustryMulti__c, Industry_SubType__c, No_of_Employees__c, address fields (Address1__c, City__c, State__c, Country__c, Zip_Postal_Code__c, Other_City__c), Last_Status_Change_Date__c, CreatedByld, CreatedDate
Account	Standard fields plus the custom fields in section 5.3, the custom Billing_ and Shipping_ address fields, Owner, Applicable_Pricebook__c (still queried even though pricebook logic is commented out), Designation__c and others read by <code>getAccount</code>

Object	Fields and notes
Contact	Salutation, FirstName, LastName, Title, Email, Phone, MobilePhone, AccountId, Department__c
Static resource	

7. Test Class: LeadConvertClsTest

`@testSetup` creates a placeholder Account named **NEW** (which makes `IsNew_Account__c` true for leads linked to it), a fully populated existing Account and a Contact, a lead for the new-Account path, a lead for the existing-Account path, and an already-converted lead.

Test	Checks
testGetLeadData_Success / _AlreadyConverted	Lead is returned; converted lead throws
testGetAccount / testGetContact	Both getters return a record
testConvertLead_NewAccount	Account and Contact created, lead marked converted, billing address mapped
testConvertLead_ExistingAccount	Existing Account used; Contact created against it with names from the lead
testConvertLead_ExistingAccount_ContactFields	Contact first name, last name, salutation, title and Account link
testConvertLead_AlreadyConverted	Returns an error message
testConvertLead_GeneralException	Null lead Id returns an error message
testConvertLead_MissingExistingAccount	Lead without Existing_Account__c returns an error message
testConvertLead_NoType	Conversion works when Type__c is blank
testConvertLead_AddressMapping	Billing and shipping address mapped
testConvertLead_LeadFieldMapping	Source, subject, note, OEM, type, classification and Contact fields mapped
testConvertLead_ContactFields	Contact linked to the new Account with email and phone
testConversionResult_DataVerification	All result fields populated
testConvertLead_BulkScenario	Five leads converted one by one with unique Account names

The test data depends on org-specific picklist values (for example Dealer_Assigned__c "Jai Balaji-Coimbatore", New_OEM__c "Bajaj Auto", Type__c "Local WB") and Account `Status__c` = `Approved`. Much of the old Sample/Quote/pricebook test code is commented out.

8. Deployment and Access

- **Deploy:** the LWC bundle, `LeadConvertCls`, `LeadConvertClsTest` and the static resource `img`, together with the Lead__c, Account and Contact fields listed in section 6.
- **Expose the component** as a quick action (or place it on a page) for Lead__c. Check the quick action and page layout in the org, because they are not part of the files supplied.

- **Access:** users need Apex class access to `LeadConvertCls`, create access to Account and Contact, edit access to `Lead__c` (`Is_Converted__c` is updated), and access to the fields used.
- **Understand the "NEW" account convention:** to start a new-Account conversion, the lead appears to point `Existing_Account__c` at an Account named NEW. Confirm this with the team, as the formula is not in the supplied files.
- **Tests:** run `LeadConvertClsTest` and check coverage.

9. Observations and Recommendations

- **Contact Name input has no effect.** The user must enter a Contact Name for new Accounts, but Apex ignores `conName` and uses the lead first and last name. Either use the typed value or remove the input.
- **Duplicates are allowed by design.** Duplicate rules are bypassed for Account and Contact. The existing-Account path creates a **new Contact every time**, even if the same person already exists.
- **No sharing keyword.** `LeadConvertCls` has no `with sharing` and the queries do not use `WITH USER_MODE`. Methods such as `getAccount` and `getContact` return data for any Id passed in. Add `with sharing` and enforce access, or remove the unused getters.
- **Unused code.** `getAccount`, `getContact`, several LWC fields and getters, unused CSS and a lot of commented-out Sample/Quote/pricebook code remain in Apex, LWC, HTML and tests. Source control keeps the history, so the dead code could be removed.
- **Logging.** Apex has many `System.debug` calls and the LWC logs the full lead (`JSON.stringify(result)`) to the browser console. Remove these, as they include personal details.
- **Conversion marker.** A lead is marked converted only by `Is_Converted__c = true`. The lead status is not changed, and the lead does not store the created Account or Contact Ids. The Account stores the link back through `Lead_Name__c` and `LeadName__c` (both are set; confirm both are needed).
- **Not bulkified.** `convertLead` handles one lead per call, which suits the UI. The "bulk" test simply calls it five times.
- **Account fields still queried.** `Applicable_Pricebook__c` is still selected, so the field must exist even though pricebook logic is off.
- **Page reload.** `handleCloseOppClick` reloads the whole page after conversion.

10. Troubleshooting

Symptom	Likely cause
"Unable to Load Lead" box	Lead already converted, user lacks access, or the lead query failed (read the message)
Convert button greyed out	A load error is set, or a conversion is running
"Account Name is required" / "Contact Name is required"	Blank inputs on the new-Account path
"An Existing Account must be selected"	Lead is not new and <code>Existing_Account__c</code> is blank

Symptom	Likely cause
"Conversion Error" toast	Validation rule, required field, or another DML error; all changes were rolled back
Contact name differs from what was typed	Apex uses the lead First_Name__c and Last_Name__c (see section 9)
Extra Contact on the existing Account	The existing-Account path always creates a new Contact
Image missing on the result screen	Static resource <code>img</code> is missing in the org